Module 0 Homework 1 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 99,990 of 99,990 points. 100%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1

Fill in the blanks:

$1 - \frac{1}{2}$ = ___,

$\frac{1}{2} - \frac{1}{3}$ = ___, and

$\frac{1}{3} - \frac{1}{4}$ = ___.

Your Answer	$\frac{1}{2}$; $\frac{1}{6}$; $\frac{1}{12}$
Correct Answer	$\frac{1}{2}$; $\frac{1}{6}$; $\frac{1}{12}$

Explanation

Did you notice a pattern? There's a nice trick here. Do you see that the answer is always 1 divided by the product of the denominators? That's because the numerators are 1, and the denominators differ by exactly 1.

To see why this is true: When you subtract two fractions of the form $(\frac{1}{a} - \frac{1}{b})$, you can make a common denominator of ab. The fractions become $\frac{b}{ab} - \frac{a}{ab}$, which is $\frac{b-a}{ab}$.

And, since the denominators differ by 1, with the second denominator b as the larger one, this is just $\frac{1}{ab}$. The denominator is the product of the denominators!

2

Fill in the blanks:

$7 \times 7 - 8 \times 6$ = ___,

$7 \times 7 - 9 \times 5$ = ___, and

$7 \times 7 - 10 \times 4$ = ___.

Your Answer	$1; 4; 9$
Correct Answer	$1; 4; 9$

Explanation

This looks like a standard set of multiplication and subtraction problems. Do you notice a pattern? Here's a hint as to why it is so nice:

You can rewrite 10×4 as $(7 + 3) \times (7 - 3)$ and expand the brackets to:

$$7 \times 7 - 7 \times 3 + 3 \times 7 - 3 \times 3$$

Now you can see why 10×4 is exactly 3^2 less than 7×7.

3 Evaluate $(-8 + 3 \times 5) \times (5 + 4 \times 8) \times (3 \times 2 - 6)$.

Your Answer	0
Correct Answer	0

Explanation

The last bracket evaluates to $3 \times 2 - 6 = 0$. Since the three brackets are multiplied together and one of them is 0, the whole result is 0, and that's the answer!

When you see a long arithmetic question like this, there is usually a trick. One strategy when you see a big multiplication is to quickly check whether any of the factors simplifies to 0, because then you don't need to do any more work. Sometimes that bracket is towards the end, to trip up people who only calculate from left to right.

If you had tried to calculate each bracket, you would have found yourself multiplying together 7×37 first, because those are the first two brackets. That would take a long time, which turns out to be unnecessary because you get to multiply by 0 in the third bracket.

4 Two distinct even positive integers sum to 8. Determine the larger of these two integers.

Your Answer	6
Correct Answer	6

Explanation

The only two possible pairs of even positive integers are $(4, 4)$ and $(2, 6)$. The first pair does not have two distinct positive integers.

5

Suppose I had a bag of marbles, with many colors, and I wanted to find the probability that if I randomly chose a marble (with all marbles having the same chance of being chosen), I would get a red one or a green one. In order to do so, I would need to find _____ over _____.

Your Answer	The total number of red and green marbles; the total number of marbles
Correct Answer	The total number of red and green marbles; the total number of marbles

Explanation

Probability is the number of favorable (good) outcomes divided by the total number of outcomes.

6 Evaluate $29 \times 29 - 33 \times 25$.

Your Answer	16
Correct Answer	16

Explanation

Sketch. Use the expansion of

$$(a+b)(a-b) = a^2 - b^2$$

on 33×25 with $a = 29$ and $b = 4$.

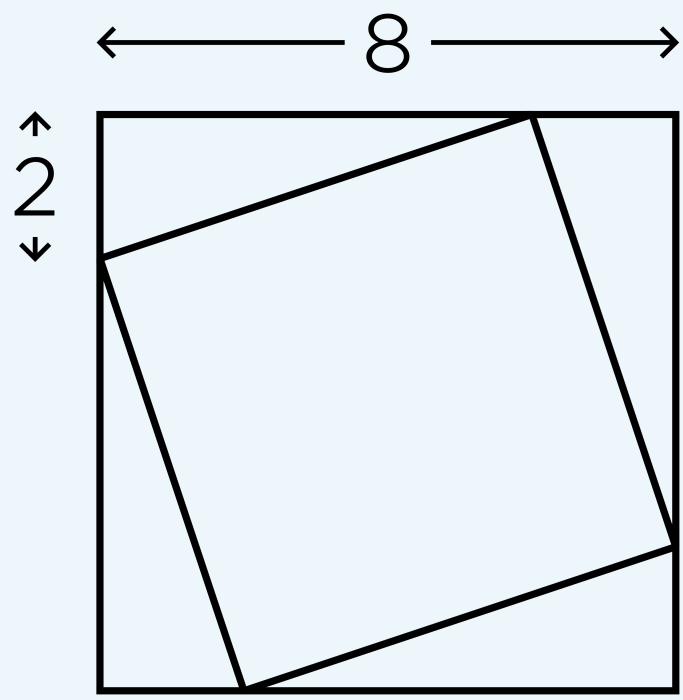
Solution. One solution is to multiply everything out and subtract, but that is tedious, and you can easily make a mistake. When you see a question like this, it's better to first ask whether there is some reason for choosing the numbers you see, because problem authors typically don't just want to give you random arithmetic to calculate. The numerical coincidence here is that 33 and 25 are both 4 away from 29. This suggests expanding brackets:

$$\begin{aligned} 33 \times 25 &= (29+4)(29-4) \\ &= 29 \times 29 - 29 \times 4 + 4 \times 29 - 4 \times 4 \\ &= 29 \times 29 - 4 \times 4. \end{aligned}$$

Since we're subtracting this from 29×29, the answer is just $4 \times 4 = 16$.

This is an example of using $(a+b)(a-b) = a^2 - b^2$ to simplify arithmetic.

7



What is the area of the inner square, if the four right-angled triangles are the same?

Your Answer	40
Correct Answer	40

Explanation

Explanation:

Solution 1. Each right triangle has legs (short sides) of length 2 and $8-2=6$. Therefore, each of the four triangles has an area of $(2\times6)/2=6$. Then, the area of the inner square is equal to the area of the big square minus the area of the four triangles. This gives an answer of:

$$8\times8-4\times6=64-24=40.$$

Solution 2. Each right triangle has legs of length 2 and 6, so by the Pythagorean Theorem, if c is the length of the hypotenuse (longest side) of the right triangle, then:

$$c^2=2^2+6^2=4+36=40.$$

Since c is also the side length of the inner square, the area of the inner square is also c^2, which is 40.

8 Albert, Beth, Carl, and Denise are trying to sit in a row of 4 seats in a movie theater.

How many ways can they sit down?

Your Answer	24
Correct Answer	24

Explanation

There are 4 choices for who sits in the first seat in the row, always 3 remaining choices for who sits in the second seat, always 2 remaining choices for who is in the third seat, and always 1 remaining choice for who is in the last seat, so the answer is $4 \times 3 \times 2 \times 1 = 24$.

9 Three faces of a rectangular box have areas 24, 20, and 30. What is the volume of the box?

✓	**Your Answer**	120
	Correct Answer	120

Explanation

Sketch. Find the prime factorization of $24 \times 20 \times 30$, and calculate its square root.

Solution. One way to solve this problem is through guess and check, but there are many possibilities to check because all of 24, 20, and 30 have many factors. The class video showed another trick, which was that if the three side lengths were a, b, and c, then 24, 20, and 30 were (in some order) ab, bc, and ac. So, if you multiply them all together, you would get exactly

$$24 \times 20 \times 30 = ab \times bc \times ac = (abc)^2,$$

which is exactly the volume squared!

The prime factorizations are:

$$\begin{aligned} 24 &= 2 \times 2 \times 2 \times 3 \\ 20 &= 2 \times 2 \times 5 \\ 30 &= 2 \times 3 \times 5 \end{aligned}$$

Therefore, the product $24 \times 20 \times 30$ is exactly:

$$\begin{aligned} &24 \times 20 \times 30 \\ &= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 \\ &= (2 \times 2 \times 2 \times 3 \times 5)^2. \end{aligned}$$

The square root of this is equal to: $2 \times 2 \times 2 \times 3 \times 5 = 120$.

10 A rectangle with integer side lengths has an area of 12. What is the greatest possible perimeter of this rectangle?

✓	**Your Answer**	26
	Correct Answer	26

Explanation

Sketch. While considering all pairs of integers that multiply to 12, observe that the perimeter is larger when the numbers are farther apart.

Solution. Since the area is 12 and side lengths are integers, we only need to consider pairs of positive integers that multiply to 12. One way to list these out is to go through the numbers $1, 2, 3, \ldots$ in order, and find out what they need to multiply with to make 12. Our first few are:

$$\begin{aligned} &1\times 12, \\ &2\times 6, \\ &3\times 4, \text{ and} \\ &4\times 3. \end{aligned}$$

Observe that when we start off, the second number 12 is bigger than the first number 1, but the second number becomes smaller than the first number at 4×3. That is how we know we can stop: we've already seen 3×4 in the list, and everything else we'd find would be a repeat of a previous product, but reversed.

The perimeter would be exactly twice the sum of length and width. Looking at the list, the largest sum of factors is $1 + 12 = 13$, corresponding to a perimeter of $13\times 2 = 26$, which is the answer.

It turns out that in general, if the product of two positive numbers is a fixed number (like 12), their sum gets larger when they move farther apart. You can see that in our problem too:

$$3 + 4 < 2 + 6 < 1 + 12$$

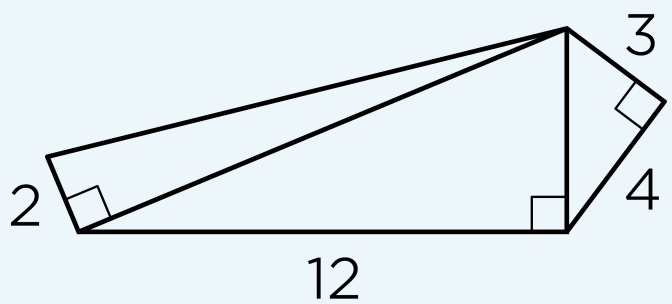
So, another way to do the problem if you knew that in advance would be to jump straight to the case where they are the farthest apart: 1 and 12 correspond to a perimeter of $2\times(1 + 12) = 26$.

Earned 4 of 7 points. 57%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11



What is the area of this pentagon, created from three right triangles?

✕	**Your Answer**	30
	Correct Answer	49

Explanation

Sketch. Repeatedly use the Pythagorean Theorem starting from the triangle with legs 3 and 4, and calculate all of the legs of all 3 right triangles. Then add up the areas of the triangles.

Solution. The total area is the sum of the areas of the 3 triangles. The smallest triangle has legs 3 and 4, and so by the Pythagorean Theorem, its hypotenuse (longest side) is 5. Using the formula $\frac{1}{2} \times \text{base} \times \text{height}$ for the area of a triangle, this smallest triangle has area:

$$\frac{1}{2} \times 3 \times 4 = 6.$$

That hypotenuse is also a leg of the biggest triangle. The other leg of that biggest triangle is 12. Suppose that its hypotenuse is c. The Pythagorean Theorem says that:

$$c^2 = 5^2 + 12^2 = 25 + 144 = 169,$$

and so c is the square root of 169, which is 13. The area of this biggest triangle is:

$$\frac{1}{2} \times 5 \times 12 = 30.$$

We actually don't need to calculate the hypotenuse of the skinny triangle, because we already have its legs (2 and 13). Its area is:

$$\frac{1}{2} \times 2 \times 13 = 13.$$

Adding together the areas of the three triangles, our total area for the pentagon is:

$$6 + 30 + 13 = 49.$$

12 How many prime numbers are divisible by 2?

Your Answer	1
Correct Answer	1

Explanation

We need to count how many multiples of 2 are also prime numbers. At first, it seems like none of them can be prime because they are all 2 times something. The only strange cases are $2 \times 0 = 0$ and $2 \times 1 = 2$, since everything from 2×2 onward clearly is not prime. Remember that 0 is not a prime number, and so 2 is the only prime number that is divisible by 2.

An easy way to remember that 0 and 1 are not prime numbers is to see that prime numbers are those integers that are useful in giving every positive integer a *unique* prime factorization. So, 0 would be useless for this purpose because you can never write a positive integer as 0 times anything. Similarly, 1 is useless for this purpose because you can multiply by as many more 1's as you want, and still get the same answer.

13 Albert, Beth, Carl and Denise are trying to sit in a row of 4 seats in a movie theater. If Albert doesn't want to sit on either of the ends, how many ways can they sit down?

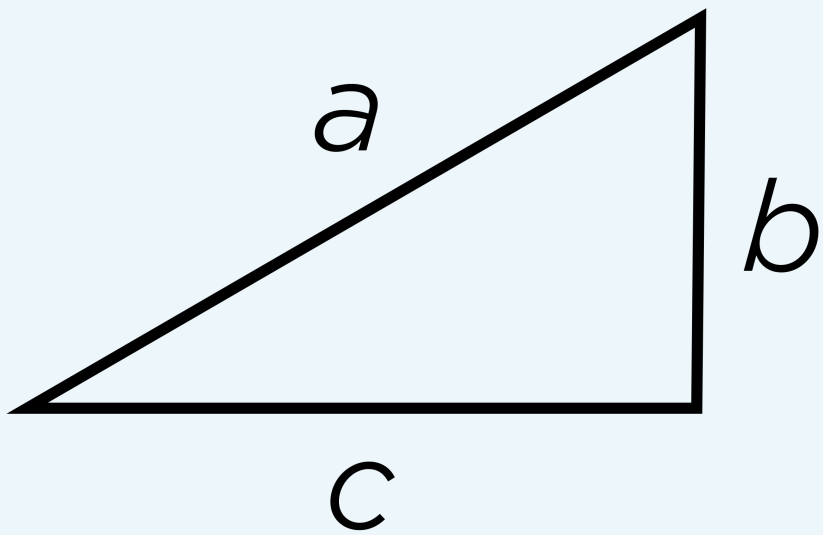
Your Answer	12
Correct Answer	12

Explanation

Sketch. Have them choose seats in alphabetical order. Albert has 2 choices in the middle. Beth always has 3 choices left, Carl then always has 2 choices left, and Denise then always has 1 choice left. Multiply: $2 \times 3 \times 2 \times 1 = 12$.

Solution. We can count the number of arrangements by supposing that Albert comes to pick his seat first, and then Beth, Carl, and Denise pick their seats in that order. Since Albert must avoid the two seats at the ends, Albert has 2 choices. Regardless of where Albert chooses to sit, Beth always has 3 choices (the other 3 seats). It is very important that this is "regardless" of Albert's choice, because that lets us multiply the numbers of choices together (if we were to draw a tree diagram, each of the 2 branches corresponding to Albert's choices would branch into exactly 3 choices for Beth). Still regardless of the first two choices, Carl always has 2 choices, because there are 2 seats left. Denise then always has 1 choice (the last remaining seat). Each set of choices corresponds to a distinct seating arrangement, and so the answer is $2 \times 3 \times 2 \times 1 = 12$.

14 Which of the following is true by the Pythagorean Theorem?



Your Answer	$b^2 + c^2 = a^2$
Correct Answer	$b^2 + c^2 = a^2$

Explanation

This is a trick question! The Pythagorean Theorem tells us that the sum of the squares of the two side lengths next to the right angle is equal to the square of the third side length. In this question's diagram, the two sides adjacent to the right angle are *b* and *c*, and the third side is *a*. Thus, the Pythagorean Theorem tells us that:

$b^2 + c^2 = a^2$

15 How many 2-digit numbers do not contain the digit 1?

Your Answer	73
Correct Answer	72

Explanation

Sketch. There are 8 choices for the tens digit and 9 choices for the ones digit. Multiply!

Solution 1. The tens digit can be anything from 2 through 9 inclusive (8 choices), because a 2-digit number cannot have 0 in the tens digit. Always be careful to treat the leftmost digit carefully, and remember to avoid 0 if the number of digits is specified in the problem. Regardless of what is placed in the tens digit, the ones digit can be any of the 10 digits except for 0, so it always has 9 choices. Multiplying, the answer is $8 \times 9 = 72$.

Solution 2. It is also possible to use complementary counting. The total number of 2-digit numbers (from 10 through 99) is $99 - 10 + 1 = 90$. The 2-digit numbers that contain the digit 1 in the tens place are:

$$10, 11, 12, 13, 14, 15, 16, 17, 18, 19.$$

There are 10 of them, which makes sense because the units digit can go from 0 through 9. The remaining 2-digit numbers that contain the digit 1 in the ones place are:

$$21, 31, 41, 51, 61, 71, 81, 91.$$

There are 8 of them, because the tens digit can range from 2 through 9. So, there is a total of $10 + 8 = 18$ of the numbers that contain the digit 1. Subtracting from the total, we get $90 - 18 = 72$.

16 How many 2-digit integers can we make such that the 2 digits are strictly increasing from left to right? For example, 12 and 36 would count, but 11 and 31 would not count.

Your Answer	36
Correct Answer	36

Explanation

Sketch. Every choice of 2 distinct digits from 1 through 9 corresponds to a 2-digit number that works. Alternatively, look for a pattern based on how many numbers work for each value of the tens digit.

Solution 1. Since the tens digit is strictly less than the ones digit, each such 2-digit number corresponds to a way of choosing 2 digits from 1 through 9, without thinking about order. For example, 36 corresponds to choosing the digits 3 and 6 in any order. Since there are 9 digits to choose from, we have learned that this is called "9 choose 2." Recall that this is calculated by taking 9×8 and dividing by 2×1, because there are 9 ways to choose the first item, and then 8 ways to choose the second item, but there are 2×1 ways to rearrange the order in which we chose the 2 items (since order does not matter). So, the answer is

$$\frac{9\times 8}{2\times 1} = 36.$$

Solution 2. If you don't see a clever trick, you can always start by systematically listing cases and looking for a pattern. If the tens digit is 1, then the ones digit can be anything from 2 through 9, which gives 8 options:

$$12, 13, 14, 15, 16, 17, 18, 19.$$

If the tens digit is 2, then the ones digit can be anything from 3 through 9, which gives 7 options:

$$23, 24, 25, 26, 27, 28, 29.$$

Following the pattern, if the tens digit is 3, then there are 6 options. Always check the last term in the pattern: if the tens digit is 8, then there is 1 option (the number 89). So, the answer is:

$$8+7+6+5+4+3+2+1.$$

You can add up this sequence by seeing that there are 8 terms, and with average term equal to the average $\frac{8+1}{2}$ of the first and last term because they are equally spaced. So, the

answer is the product of the number of terms and their average, which is:

$$8 \times \frac{8+1}{2} = 36.$$

17 Suppose we want to find the number of 3 digit numbers which contain the digit 9 at least once. We can do this by subtracting _____ from _____ (called *complementary counting*).

Your Answer	The total number of 3-digit numbers; the number of 3-digit numbers which don't contain the digit 9
Correct Answer	The number of 3-digit numbers which don't contain the digit 9 ; the total number of 3-digit numbers

Explanation

Make sure to read the question carefully! The idea of complementary counting is to subtract the set of numbers that we do not want from some "total" set of numbers. In this case, we don't want the numbers that don't contain 9, and our total set is all of the 3-digit numbers.

Earned 3 of 3 points. 100%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 How many 3-digit integers have the property that the product of their digits is 30?

Your Answer	12
Correct Answer	12

Explanation

Explanation:
Sketch. The only possible sets of digits are $(1, 5, 6)$ and $(2, 3, 5)$ and each of them has $3 \times 2 \times 1 = 6$ rearrangements into 3-digit numbers.

Solution. The first step of this problem is to find triples of 3 single-digit numbers which multiply to 30. The prime factorization of $30 = 2 \times 3 \times 5$. Let's find all single-digit numbers that are factors of 30. Using the prime factorization or manually checking, we see that the only such digits are:

$$1, 2, 3, 5, 6.$$

Since 30 is a multiple of 5 and only one of these digits can contribute a multiple of 5, we must use 5 as one of the digits. Thus, it only remains to find pairs of digits which multiply to 6. We can easily find that the only possibilities are 1×6 and 2×3.

Therefore, the only possible products are $5 \times 1 \times 6$ and $5 \times 2 \times 3$. Each of those two possibilities can be rearranged in exactly $3 \times 2 \times 1 = 6$ ways (given a set of 3 distinct digits, there are 3 ways to pick which digit goes first, then 2 ways to pick which digit goes next, and then 1 way to pick the last digit). So, the answer is $2 \times 6 = 12$.

19 Alina and Becky are playing a game where they flip a fair coin 3 times. Alina wins if the number of heads is even, and Becky wins otherwise. What is the percent chance that Alina wins, rounded to the nearest percent? (If the answer is 31.4 you would type in 31.)

Your Answer	50
Correct Answer	50

Explanation

Sketch. Either use brute force to list all 2^3 outcomes and count how many are good, or observe that regardless of the outcomes of the first two flips, Alina always has a 50% chance of exactly reversing the evenness/oddness (called "parity") with her third flip.

Solution 1. Since 3 times is not too many, brute force is actually possible. There will be exactly $2\times2\times2 = 8$ outcomes in total because there are 2 choices for the outcome of the first coin, 2 choices for the second coin, and 2 choices for the third coin. The 8 outcomes can be systematically listed as:

$$\begin{aligned}
&HHH\text{: odd number of heads}\\
&HHT\text{: even number of heads}\\
&HTH\text{: even}\\
&HTT\text{: odd}\\
&THH\text{: even}\\
&THT\text{: odd}\\
&TTH\text{: odd}\\
&TTT\text{: even}
\end{aligned}$$

Since 4 of the outcomes are favorable (even number of heads), and there are 8 total outcomes, the probability is $\frac{4}{8} = 50\%$.

Solution 2. An amazingly clever solution is to notice that no matter what happens in the first two flips, the last flip will either be tails (keeping the parity of the total number of heads) or heads (changing the parity of the total number of heads). The "parity" of a number is its evenness or oddness.

Consider the two possible cases before the last flip: if there was an odd number of heads before the last flip, then there is a 50% chance that the last flip changes the parity (to let Alina win); on the other hand, if there was an even number of heads before the last flip, then there is also a 50% chance that the last flip keeps the parity (to let Alina win). In both cases, there is a 50% chance that Alina wins, and so that is the answer.

Observe that this solution works even if the coin is flipped 100 times, showing that there is a 50% chance that at the end, there is an even number of heads. It would be impossible to list all possibilities with that many flips!

20 There are 6 empty chairs arranged in a circle. Three people arrive, and randomly choose places to sit, where all possibilities are equally likely. What is the percent chance that no two people sit next to each other, rounded to the nearest percent? (If the answer is 31.4%, you would type in 31.)

✓	**Your Answer**	10
	Correct Answer	10

Explanation

Explanation:
Sketch. Either use the multiplication principle to count and divide possibilities, or observe that the first person's seat doesn't matter by symmetry, but once the first person sits down, there are exactly 2 seats that the other two people must sit in.

Solution 1. The only possible way that three people can sit down in six seats without being next to each other is by sitting every other seat.

The total number of outcomes is $6 \times 5 \times 4 = 120$ because the first person has 6 choices of where to sit, the second person has 5 choices left, and the third person has 4 choices left.

Now, in order to count the number of favorable outcomes, note that the first person has 6 choices of seats, all of which can become acceptable seating arrangements. Since the other two people must then sit every other seat, once the first person sits down, there are two specific seats which the other two people must sit in. There are always 2 ways for them to arrange themselves in those two seats, and so the number of favorable outcomes is $6 \times 2 = 12$. Therefore, the probability is equal to $\frac{12}{120} = 10\%$.

Solution 2. Another way to think about this is to observe that no matter where the first person sits down, there are now 2 specific seats which must attract the other two people. When the second person comes, he sees 5 seats (since 1 is already occupied), and he must sit in one of the 2 seats that have been identified, which has a probability of $\frac{2}{5}$.

When the third person comes, she sees 4 seats (since 2 are already occupied), and she must sit in the other of the 2 seats that have been identified, which has a probability of $\frac{1}{4}$. Multiplying the probabilities, the answer is $\frac{2}{5} \times \frac{1}{4} = 10\%$.

Summary

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